

Chapter 3: Exponents, Radicals & Logarithms

Laws of Exponents, Radical Simplification, Logarithmic Properties & Exponential Equations

3.1 Laws of Exponents & Radicals

Exponents denote repeated multiplication. Rational exponents link directly to radical expressions: $a^{m/n} = n\sqrt[n]{a^m}$.

Exponent Laws

- *Product Rule:* $a^m \cdot a^n = a^{m+n}$
- *Quotient Rule:* $a^m / a^n = a^{m-n}$
- *Power of Power:* $(a^m)^n = a^{m \cdot n}$
- *Negative Exponent:* $a^{-n} = 1 / a^n$
- *Zero Exponent:* $a^0 = 1$ ($a \neq 0$)

Radical Properties

- $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$
- $\sqrt[n]{a/b} = \sqrt[n]{a} / \sqrt[n]{b}$
- *Rationalizing Denominators:*
 $1 / \sqrt[n]{a} = \sqrt[n]{a} / a$
 $1 / (\sqrt[n]{a} + \sqrt[n]{b}) = (\sqrt[n]{a} - \sqrt[n]{b}) / (a - b)$

3.2 Fundamental Logarithmic Properties

A logarithm is the inverse operation to exponentiation: $\log_b(x) = y \Rightarrow b^y = x$ (where $b > 0$, $b \neq 1$, $x > 0$).

- *Product Rule:* $\log_b(MN) = \log_b(M) + \log_b(N)$
- *Quotient Rule:* $\log_b(M/N) = \log_b(M) - \log_b(N)$
- *Power Rule:* $\log_b(M^k) = k \cdot \log_b(M)$
- *Change of Base Formula:* $\log_b(x) = \ln(x) / \ln(b) = \log(x) / \log(b)$

Worked Example 3.1: Logarithmic Expansion

Problem: Expand $\log_2(8x^3/y^2)$ into individual logarithmic terms.

Solution: Apply quotient and product rules:

$$\log_2(8) + \log_2(x^3) - \log_2(y^2)$$

Since $8 = 2^3$, $\log_2(8) = 3$. Apply power rule:

$$= 3 + 3 \log_2(x) - 2 \log_2(y).$$

3.3 Solving Exponential & Logarithmic Equations

To solve exponential equations, take the natural logarithm ($\ln$) of both sides or equate bases. For logarithmic equations, combine terms into a single log and convert to exponential form.

Worked Example 3.2: Exponential Equation

Problem: Solve for x : $5^{2x-1} = 27$.

Solution: Take natural logarithms on both sides:

$$\ln(5^{2x-1}) = \ln(27) \Rightarrow (2x-1) \ln(5) = \ln(27)$$

$$2x-1 = \ln(27) / \ln(5) \Rightarrow 2x = 1 + [\ln(27) / \ln(5)]$$

$$x = 1/2 + \ln(27) / [2 \ln(5)] \approx 1.523.$$

Worked Example 3.3: Logarithmic Equation

Problem: Solve for x : $\log(x) + \log(x-3) = 1$ (Base 10).

Solution: Combine using the product rule:

$$\log[x(x-3)] = 1 \Rightarrow \text{Convert to exponential form: } x(x-3) = 10^1$$

$$x^2 - 3x - 10 = 0 \Rightarrow (x-5)(x+2) = 0 \Rightarrow x = 5 \text{ or } x = -2.$$

Check extraneous solutions: $\log(-2)$ is undefined. Therefore, $x = 5$.

Chapter 3 Review & Practice Problems

1. Simplify completely: $(2x^3y^{-2})^3 / (4x^{-1}y^4)$.
2. Rationalize the denominator of: $(3 + \sqrt{2}) / (5 - \sqrt{2})$.
3. Condense into a single logarithmic expression: $2 \ln(x) + 3 \ln(y) - 1/2 \ln(z)$.
4. Solve for x : $e^{3x+1} = 15$.
5. Solve the logarithmic equation: $\log_2(x+1) - \log_2(x-1) = 3$.

